CINCINNATI REDS Z-TEST LAB

Student Instructions

Module 8 / Prep 8 · One-Sample Z-Test · Companion to *reds_ztest.xlsx*

The Setup

You work in the front office of the Cincinnati Reds. League-wide (MLB) averages are public knowledge — those are your POPULATION figures. You've collected a SAMPLE of Reds data, and your job is to answer one question three times: is the Reds number REALLY different from the league, or is the gap just luck of the draw?

Before You Begin

- Open *reds_ztest.xlsx*. You'll see five tabs along the bottom. Begin on "Start Here," then work the tabs left to right.
- Keep your Module 8 notes open beside you — the workbook follows them exactly (same four blocks, same formulas, same logic).
- Save your file often. Type Excel formulas or use cell references

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Deleted: (don't paste numbers); the lab is checking that you can build the test, not just report an answer.

This cell looks like…	…which means
Yellow fill	You fill it in. Type the formula or your answer here.
Blue text	A given input (μ_0, σ, α). You may change it to explore, but undo afterward.
Black text	A formula that calculates automatically. Click it to see how it works.
Green italics	A hint showing the Excel formula you need for the cell beside it.

How You'll Learn It — Four Stages

The tabs are sequenced so you experience a z-test before you formalize it, then build one on your own. Work them in this order:

Stage	What you do
① **Experience**	Tab "1. Attendance" is fully built and live. Run it and poke at it to get a feel for a complete z-test.
② **Reflect**	Tab "2. Game Time" is half-built. You finish it, then notice why it lands on a different decision than Tab 1.
③ **Conceptualize**	On "Reflect & Apply," put the rules in your own words — the doing becomes principles.
④ **Experiment**	Tab "3. Ticket Price" is an empty skeleton. Predict the result, then build the whole test yourself.

Part 1 — Tab “1. Attendance” (explore the built model)

1. Read the scenario like a detective. Find and mark the five facts: the population mean μ_0, the population standard deviation σ, the sample size n, the significance level α, and the claim word.
2. This tab is already built. Click through Blocks 1–4 and read the green hints — see how each cell is calculated.
3. Experiment: change a blue input (try lowering σ, or raising μ_0) and watch the test statistic, p-value, and decision update. Then press **Ctrl+Z** to restore it.
4. Make sure you can say out loud WHY this test rejects the null. Then move on.

Part 2 — Tab “2. Game Time” (complete the test)

Block 1 is done for you. You fill the yellow cells in Blocks 2–4 and the decision, using the green hints and your notes.

1. Confirm the hypotheses and the tail at the top (the claim word “less / shorter” makes it left-tailed).
2. Block 2 – cutoffs. For a left tail, the probability cell is α, the critical z is `=NORM.S.INV(prob)` — **no “1−” on the left.**
3. Block 3 – test statistic. Compute z, then the left-tail p-value with `=NORM.S.DIST(z,TRUE)`.
4. Block 4 – the 95% confidence interval (margin of error, then lower and upper bounds).
5. Decision and conclusion: compare p to α, then write the result in symbols and in plain English.

Part 3 — Tab “Reflect & Apply” (put it into words)

Answer the prompts in the yellow cells. They move from what you noticed (Reflect), to the rules behind it (Conceptualize), to pushing the ideas further (Experiment). Full sentences — this is where the statistics actually stick.

Part 4 — Tab “3. Ticket Price” (build it yourself)

This tab is an empty skeleton — just labels and the data. Build the entire test from your notes and Tab 1 as your model.

1. Build Block 1 (use =AVERAGE and =COUNT on the data column), then Blocks 2–4.
2. Mind the two-tailed differences: the probability cell is **α/2**, you need BOTH a left and a right critical z, the p-value depends on the sign of z, and you compare p to **α/2**.
3. Write the decision, the conclusion (state the direction — higher or lower), and the confidence-interval sentence.

Deleted: <#>**Predict first.** Before any formulas, write your guess (reject or fail to reject?) in the “Predict First” box.¶

Deleted: <#>Check your prediction against the result. Were you right? Why or why not?¶

Quick Reference

The four blocks (every tab is the same)

Block	What you compute	Excel
1	Comparison data: μ_0, $\bar{x}$, n, σ, SE	`=AVERAGE(…)` `=COUNT(…)` `=σ/SQRT(n)`
2	Critical cutoffs: prob, critical z, critical $\bar{x}$	`=NORM.S.INV(…)` `x̄crit = z×SE+μ₀`
3	Test statistic: z and its p-value	`z=(x̄−μ₀)/SE` `=NORM.S.DIST(z,TRUE)`
4	95% confidence interval: MOE, bounds	`MOE=NORM.S.INV(1−α/2)×SE; x̄±MOE`

What changes with each tail

Step / cell	Right	Left	Two-tailed
Probability cell	α	α	α / 2
Left critical z	—	NORM.S.INV(prob)	NORM.S.INV(prob)
Right critical z	NORM.S.INV(1−prob)	—	NORM.S.INV(1−prob)
p-value	1−NORM.S.DIST(z)	NORM.S.DIST(z)	sign of z: + use right, − use left
Compare p to	α	α	α / 2

What to Turn In

- The saved workbook with
  1. Tab 2 (Game Time)
  2. Tab 3 (Ticket Price) completed
  3. Tab “Reflect & Apply” answered
- Your one-sentence takeaway from all three tests (the last Reflect prompt).

Instructor Notes

- Distribute reds_ztest.xlsx with these instructions. Delete the “Answer Key (Instructor)” tab before handing the file to students.
- Tab 1 is fully live; Tab 2 has Block 1 pre-filled with the rest blank; Tab 3 is a blank skeleton. The built/answer cells auto-write decisions and conclusions in plain English, so they are self-checking.
- **Expected outcomes:** Tab 1 Attendance → reject ($p \approx 0.8\%$, higher). Tab 2 Game Time → fail to reject ($p \approx 30\%$) — note the sample mean is below the league yet not significantly so; that contrast is the teaching point. Tab 3 Ticket Price → reject ($p \approx 0.25\%$, different/lower).
- Suggested time: 45–60 minutes; works well in pairs with the Reflect tab done individually.
- The Answer Key tab’s formulas read each tab’s data column, so you can swap in different numbers (or current real MLB figures) and the key updates automatically.
- The MLB averages used (28,000 fans; 156-minute games; $38 tickets; the listed σ values) are illustrative for the exercise, not official league statistics — update them if you want the scenarios to be literally accurate.